

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
RUMOMASI**

TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: SEVEN

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Information & Communication Literacy

Introduction

- Information literacy is the ability to find, evaluate, use and create information effectively; communication literacy is the ability to communicate through media correctly, ethically and safely. Together they are the core skills of the information age.

Finding and evaluating information

- Search strategies (keywords, filters, boolean operators AND/OR/NOT); evaluate sources by authority (who wrote it?), accuracy (is it verifiable?), currency (when?), and bias (whose view?); prefer primary, peer-reviewed and official sources to blogs and social media; and cite your sources honestly.

Ethical and safe use

- Respect copyright (use and attribute, avoid plagiarism — paraphrase and quote properly); protect privacy (personal data, passwords); verify before sharing (fake news and misinformation spread faster than corrections); think critically about influencer content and deep-fakes; and observe netiquette.

CLASS EVALUATION

1. Define information literacy.
2. What four questions evaluate a source?
3. What is plagiarism and how is it avoided?
4. State two rules of responsible sharing.

**NAME OF SCHOOL: MATER MISERICORDIAE SECONDARY SCHOOL,
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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: EIGHT

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Computer Maintenance

Introduction

- Computer maintenance is the routine care that keeps hardware and software working and safe: preventive (planned) and corrective (after a fault) maintenance.

Hardware care

- Keep the machine clean — dust the vents and fans (compressed air); clean the screen and keyboard; keep drinks and food away; protect against power surges (the UPS and surge protector); prevent overheating (good ventilation); avoid static by grounding before touching the internals; handle storage drives gently; and service the machine regularly.

Software maintenance

- Update the operating system and applications (patch vulnerabilities); run antivirus and malware scans; clean up junk files and the browser cache; defragment/optimize storage; uninstall unused programs; back up data regularly (the 3-2-1 rule: three copies, two media, one off-site); and tidy the desktop and filing.

Troubleshooting

- Step 1: identify the problem (what changed? when?); 2: state the likely cause; 3: test one fix at a time; 4: record the repair. Common problems: the machine will not power on (power, cables, battery), slow performance (too many programs, full disk, RAM), no internet (router, drivers), a virus alert, and the 'blue screen'.

CLASS EVALUATION

1. Define preventive maintenance.
2. State two hardware and two software care practices.
3. What is the 3-2-1 backup rule?
4. List the four steps of troubleshooting.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: NINE

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Career Paths in ICT

Introduction

- ICT careers are the fastest-growing employment sector: programming, networking, cybersecurity, data, design and support. This week maps the career paths and how to prepare for them.

The main careers

- **Software development** — programmers, software engineers, mobile app developers, game developers (Java, Python, JavaScript, C#); **networking** — network engineers, administrators (CCNA); **cybersecurity** — analysts, ethical hackers, security architects; **data** — data analysts, data scientists (Python, SQL, machine learning); **web and cloud** — web developers, cloud engineers (AWS, Azure); **AI and robotics** — machine learning engineers; **support** — IT support technicians, helpdesk; **other** — database administrators, UI/UX designers, multimedia and games.

Education and certification

- Formal: computer science, IT, software engineering degrees, ND/HND, the national certificates; professional certifications: CompTIA A+ (support), Cisco CCNA (networking), Microsoft and AWS certifications, and Google courses; bootcamps and online platforms (Coursera, Udemy, freeCodeCamp) give practical skills.

The Nigerian ICT economy

- Job markets: banks and telecoms, the tech hubs (Yaba, Lagos), outsourcing (call centres, data services), government (INEC, NIMC, the NITDA), freelancing (Upwork, Fiverr) and entrepreneurship (fintech — the 'Nigerian fintech wave').

CLASS EVALUATION

1. Name five ICT careers.
2. What does a network engineer do?
3. Which certification is the entry point for IT support?
4. State two ways of acquiring ICT skills outside a university.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TEN

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: ICT & Society (Digital Citizenship)

Introduction

- ICT has transformed society: work, education, business, governance and social life. Digital citizenship is the responsible, ethical and safe use of technology — and understanding technology's effects on society.

Effects of ICT

- Positive: e-learning and open education; e-government (the NIN, BVN, e-tax, e-passport); e-commerce and fintech (transfers, POS, mobile money); telemedicine; telework; social connection; access to information; and the growth of the digital economy (jobs, startups). Negative: the digital divide (urban/rural, rich/poor); job displacement and the skills gap; cybercrime (yahoo-yahoo, phishing, scams); misinformation and fake news; privacy erosion and surveillance; screen addiction; cyberbullying; and the environmental cost of e-waste.

Laws and protection

- The Cybercrime (Prohibition, Prevention, etc.) Act 2015; the Nigeria Data Protection Act 2023; the NITDA and the NCC as regulators; child online protection (parental controls); and reporting tools (the cybercrime reporting on the Police website).

The responsible digital citizen

- Verify before sharing; protect data and privacy; respect others online; avoid fraud and piracy; use technology for learning and productivity; report cybercrime; and close the divide by sharing skills.

CLASS EVALUATION

1. Define digital citizenship.
2. State four positive and four negative effects of ICT.
3. Name two Nigerian ICT laws.
4. How can you tell a fake news story before sharing it?

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: ELEVEN

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Introduction to Programming (Algorithms & Flowcharts)

Introduction

- A program is a set of instructions that a computer executes. Programming is writing those instructions in a language the computer can run; an algorithm is the step-by-step plan of the solution — the thinking before the coding.

Algorithm and pseudocode

- An algorithm is a finite, ordered set of steps that solves a problem. Properties: it must be clear, unambiguous, finite (it must end), and effective (each step does something). Pseudocode is the algorithm written in plain English-like statements: START, INPUT, PROCESS, IF...THEN...ELSE, REPEAT, OUTPUT, END.

Flowcharts

- The diagram of an algorithm: **oval** — START/STOP; **parallelogram** — input/output (READ, PRINT); **rectangle (process)** — calculation and assignment; **diamond** — decision (yes/no); **arrows** — the flow of control. Example (the average of two numbers): START → READ A, B → $S = A + B$ → $AVE = S/2$ → PRINT AVE → STOP. Example (grading): IF score ≥ 70 PRINT 'A' ELSE IF score ≥ 60 PRINT 'B' ... ELSE PRINT 'F'.

The basics of coding

- Programming languages: low-level (machine and assembly) and high-level (BASIC, Pascal, Python, Java, C++, JavaScript); a compiler translates the whole program, an interpreter line by line. Basic structures: **sequence** (one step after another), **selection** (if/else) and **iteration** (loops — for, while). First Python lines: `print('Hello')`, `name = input('Name: ')`, and a for loop that prints the numbers 1 to 10.

CLASS EVALUATION

- Define a program and an algorithm.
- State the three properties of an algorithm.
- Draw the flowchart symbols for input, decision and process.
- Write pseudocode for the average of three numbers.
- Differentiate a compiler from an interpreter.

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TERM: SECOND 2026/2027 ACADEMIC SESSION

WEEK: TWELVE

CLASS: SS3

SUBJECT: Digital Technology

TOPIC: Databases & Spreadsheets (Practical Revision)

Introduction

- Data are raw facts; information is processed data. Databases and spreadsheets are the two main tools for managing data on the computer — and both appear in the practical examinations.

Databases

- A database is an organized collection of related data. Terms: a **table (file)** — e.g. STUDENTS; a **record (row)** — one student; a **field (column)** — one item (name, class); a **primary key** — the unique field (the admission number); forms and reports; queries — questions the database answers. Database management systems (DBMS): Microsoft Access, SQL Server, MySQL; SQL is the language of databases — `SELECT * FROM STUDENTS WHERE CLASS = 'SS2'`.

Spreadsheets

- A spreadsheet is a grid of cells (columns A, B, C...; rows 1, 2, 3...) — e.g. Excel and Google Sheets. Formulas begin with '=': `=A1+B1`, `=SUM(A1:A10)`, `=AVERAGE(B1:B12)`, `=MAX` and `=MIN`, `=IF(score>=50, 'Pass', 'Fail')`, `=COUNTIF`; relative and absolute references (`$A$1`); sorting and filtering; and charts (bar, pie, line) for presentation.

Practical applications

- The school's results sheet: enter the marks; compute the totals and averages; grade with IF; rank with RANK; chart the performance of the class; print. The mini project: a database of the school's books or students, with input forms, queries (all students in SS2) and a report of the results.

CLASS EVALUATION

1. Differentiate data from information.
2. Define record, field and primary key.
3. Write a formula for the average of the cells B1 to B10.
4. What does `=IF(A1>=50,'Pass','Fail')` display if A1 is 40?
5. State two uses of a spreadsheet in a school.

